What is a DB and Why Use One?

What is a DB?

- Database: A collection of data
- Database Management System (DBMS): Software for managing databases



Why use a DB?

Three core goals:

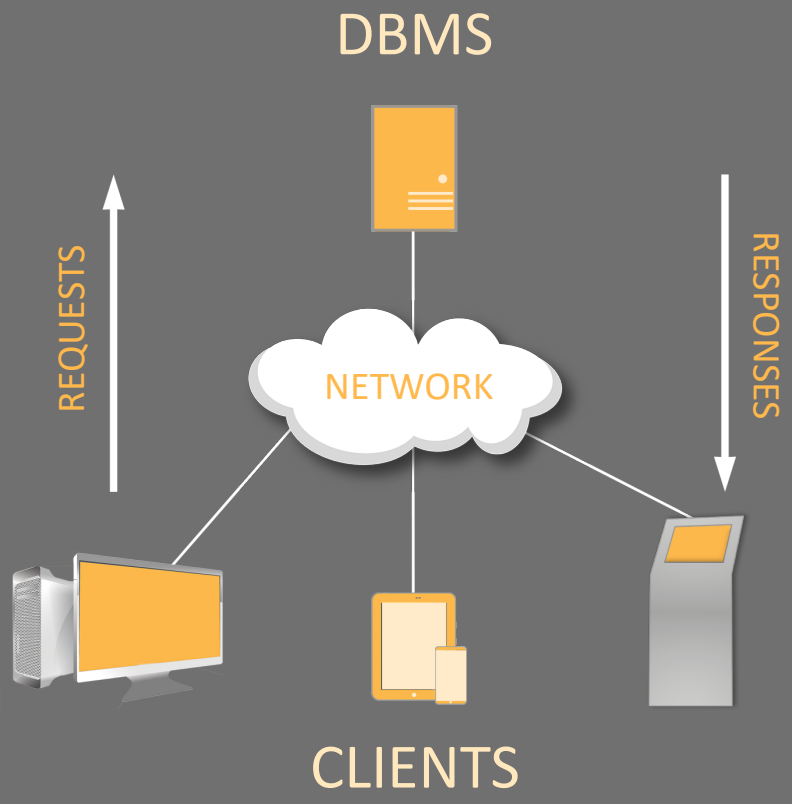
- Preserve data
- Flexible search & computation
- Independence

Plus usability and speed!

Server-based

- Typical client requests (SQL):
  - Create database
  - Add, modify, delete data
  - Select data

- Session = client login until logout
- Data persists after session.



Why not familiar tools, instead?





Many Options

Simple Data Storage
- Flat files
- Spreadsheets

Relational

T.+A.
- Oracle
- SAP
- Microsoft SQLServer

Transactional
- PostgreSQL
- MySQL

Analytical
- Cognos Powerplay
- MicroStrategy

NoSQL

Column Store
- HBase
- Cassandra

Key-Value
- Riak
- Amazon S3

Graph
- Neo4J
- OrientDB
- Giraph

Document
- MongoDB
- Couchbase

Cluster-computing framework
- Spark

Reality: Tremendous overlap. Many systems support ideas in multiple categories.